# CNNP Skill 1: Encapsulation Builder

Name: Tejaswin Amara  Roll No.: 2520090104

## 1. Install Wireshark, TShark and test tools

```
sudo apt update
sudo apt install -y wireshark tshark curl dnsutils iputils-ping
```

**Observation:** Installs the capture and test tools. Packages were already at the latest version or were upgraded during install.

```
tejaswin@DESKTOP-P4G181S:~$ sudo apt update
sudo apt install -y wireshark tshark curl dnsutils iputils-ping
[sudo: authenticate] Password:
Get:1 http://security.ubuntu.com/ubuntu resolute-security InRelease [137 kB]
Hit:2 http://archive.ubuntu.com/ubuntu resolute InRelease
Get:3 http://archive.ubuntu.com/ubuntu resolute-updates InRelease [137 kB]
Get:4 http://security.ubuntu.com/ubuntu resolute-security/main amd64 Packages [717 kB]
Hit:5 http://archive.ubuntu.com/ubuntu resolute-backports InRelease
Get:6 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 Packages [868 kB]
Get:7 http://security.ubuntu.com/ubuntu resolute-security/universe amd64 Packages [220 kB]
Get:8 http://archive.ubuntu.com/ubuntu resolute-updates/universe amd64 Packages [389 kB]
Fetched 2469 kB in 4s (653 kB/s)
142 packages can be upgraded. Run 'apt list --upgradable' to see them.
Note, selecting 'bind9-dnsutils' instead of 'dnsutils'
wireshark is already the newest version (4.6.4-1).
tshark is already the newest version (4.6.4-1).
iputils-ping is already the newest version (3:20250605-1ubuntu1).
iputils-ping set to manually installed.
Upgrading:
  curl  libcurl3t64-gnutls  libcurl4t64

Installing:
  bind9-dnsutils

Installing dependencies:
  bind9-host  bind9-libs  liblmdb0  liburcu8t64  libuv1t64

Summary:
  Upgrading: 3, Installing: 6, Removing: 0, Not Upgrading: 139
  Download size: 2870 kB
  Space needed: 5061 kB / 1024 GB available

Get:1 http://archive.ubuntu.com/ubuntu resolute/main amd64 liblmdb0 amd64 0.9.31-1build2 [49.5 kB]
Get:2 http://archive.ubuntu.com/ubuntu resolute/main amd64 liburcu8t64 amd64 0.15.6-1 [64.7 kB]
Get:3 http://archive.ubuntu.com/ubuntu resolute/main amd64 libuv1t64 amd64 1.51.0-2ubuntu1 [103 kB]
Get:4 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 bind9-libs amd64 1:9.20.24-1ubuntu0.5 [1316 kB]
Get:5 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 bind9-host amd64 1:9.20.24-1ubuntu0.5 [49.8 kB]
Get:6 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 bind9-dnsutils amd64 1:9.20.24-1ubuntu0.5 [164 kB]
Get:7 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 curl amd64 8.18.0-1ubuntu2.8 [273 kB]
Get:8 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 libcurl4t64 amd64 8.18.0-1ubuntu2.8 [430 kB]
Get:9 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 libcurl3t64-gnutls amd64 8.18.0-1ubuntu2.8 [421 kB]
Fetched 2870 kB in 11s (266 kB/s)
Selecting previously unselected package liblmdb0:amd64.
(Reading database ... 40582 files and directories currently installed.)
Preparing to unpack .../0-liblmdb0_0.9.31-1build2_amd64.deb ...
Unpacking liblmdb0:amd64 (0.9.31-1build2) ...
Selecting previously unselected package liburcu8t64:amd64.
Preparing to unpack .../1-liburcu8t64_0.15.6-1_amd64.deb ...
Unpacking liburcu8t64:amd64 (0.15.6-1) ...
Selecting previously unselected package libuv1t64:amd64.
Preparing to unpack .../2-libuv1t64_1.51.0-2ubuntu1_amd64.deb ...
Unpacking libuv1t64:amd64 (1.51.0-2ubuntu1) ...
Selecting previously unselected package bind9-libs:amd64.
Preparing to unpack .../3-bind9-libs_1%3a9.20.24-1ubuntu0.5_amd64.deb ...
Unpacking bind9-libs:amd64 (1:9.20.24-1ubuntu0.5) ...
Selecting previously unselected package bind9-host.
Preparing to unpack .../4-bind9-host_1%3a9.20.24-1ubuntu0.5_amd64.deb ...
Unpacking bind9-host (1:9.20.24-1ubuntu0.5) ...
Selecting previously unselected package bind9-dnsutils.
Preparing to unpack .../5-bind9-dnsutils_1%3a9.20.24-1ubuntu0.5_amd64.deb ...
Unpacking bind9-dnsutils (1:9.20.24-1ubuntu0.5) ...
Preparing to unpack .../6-curl_8.18.0-1ubuntu2.8_amd64.deb ...
Unpacking curl (8.18.0-1ubuntu2.8) over (8.18.0-1ubuntu2.4) ...
Preparing to unpack .../7-libcurl4t64_8.18.0-1ubuntu2.8_amd64.deb ...
Unpacking libcurl4t64:amd64 (8.18.0-1ubuntu2.8) over (8.18.0-1ubuntu2.4) ...
Preparing to unpack .../8-libcurl3t64-gnutls_8.18.0-1ubuntu2.8_amd64.deb ...
Unpacking libcurl3t64-gnutls:amd64 (8.18.0-1ubuntu2.8) over (8.18.0-1ubuntu2.4) ...
Setting up liblmdb0:amd64 (0.9.31-1build2) ...
Setting up liburcu8t64:amd64 (0.15.6-1) ...
Setting up libcurl4t64:amd64 (8.18.0-1ubuntu2.8) ...
Setting up libuv1t64:amd64 (1.51.0-2ubuntu1) ...
Setting up libcurl3t64-gnutls:amd64 (8.18.0-1ubuntu2.8) ...
Setting up curl (8.18.0-1ubuntu2.8) ...
Setting up bind9-libs:amd64 (1:9.20.24-1ubuntu0.5) ...
Setting up bind9-host (1:9.20.24-1ubuntu0.5) ...
Setting up bind9-dnsutils (1:9.20.24-1ubuntu0.5) ...
Processing triggers for man-db (2.13.1-1build1) ...
Processing triggers for libc-bin (2.43-2ubuntu2.3) ...
tejaswin@DESKTOP-P4G181S:~$
```

*Figure: apt update and install output*

## 2. Check network interfaces

```
ip -br link
sudo tshark -D
```

**Observation:** eth3 is UP, loopback0 is UP, eth0, eth1 and eth2 are DOWN. TShark lists 20 capture devices, including eth3.

```
tejaswin@DESKTOP-P4G181S:~$ ip -br link
sudo tshark -D
lo               UNKNOWN        00:00:00:00:00:00 <LOOPBACK,UP,LOWER_UP>
eth0             DOWN           82:13:16:a1:19:fe <BROADCAST,MULTICAST>
loopback0        UP             00:15:5d:e3:85:90 <BROADCAST,MULTICAST,UP,LOWER_UP>
eth1             DOWN           80:13:16:a1:19:fe <BROADCAST,MULTICAST>
eth2             DOWN           80:13:16:a1:19:ff <BROADCAST,MULTICAST>
eth3             UP             d0:37:45:2e:63:10 <BROADCAST,MULTICAST,UP,LOWER_UP>
Running as user "root" and group "root". This could be dangerous.
1. loopback0
2. eth3
3. any
4. lo (Loopback)
5. bluetooth-monitor
6. nflog
7. nfqueue
8. dbus-system
9. dbus-session
10. eth0
11. eth1
12. eth2
13. ciscodump (Cisco remote capture)
14. dpauxmon (DisplayPort AUX channel monitor capture)
15. randpkt (Random packet generator)
16. sdjournal (systemd Journal Export)
17. sshdig (SSH remote syscall capture)
18. sshdump (SSH remote capture)
19. udpdump (UDP Listener remote capture)
20. wifidump (Wi-Fi remote capture)
```

*Figure: Interface list and TShark capture devices*

## 3. Capture on eth0 without sudo (failed)

```
sudo tshark -i eth0 -w ~/sender_capture.pcapng
```

**Observation:** the capture session could not be initiated. Permission was denied on eth0.

```
tejaswin@DESKTOP-P4G181S:~$ sudo tshark -i eth0 -w ~/sender_capture.pcapng
Running as user "root" and group "root". This could be dangerous.
Capturing on 'eth0'
tshark: The capture session could not be initiated due to error getting information on pipe or socket: Permission denied.
```

*Figure: TShark permission error on eth0*

## 4. Generate traffic: curl, dig and ping

```
curl -sI https://www.google.com
nslookup google.com
ping -c 4 8.8.8.8
```

**Observation:** HTTP/2 200 response from Google, DNS answers for google.com, and ICMP echo replies from 8.8.8.8 (reply TTL 113).



*Figure: curl, dig and ping output*

## 5. Set permissions on the saved capture (failed)

```
sudo chmod 644 ~/sender_capture.pcapng
ls -lh ~/sender_capture.pcapng
```

**Observation:** 'No such file or directory'. The capture file was not written to /home/tejaswin, so chmod and ls cannot find it.

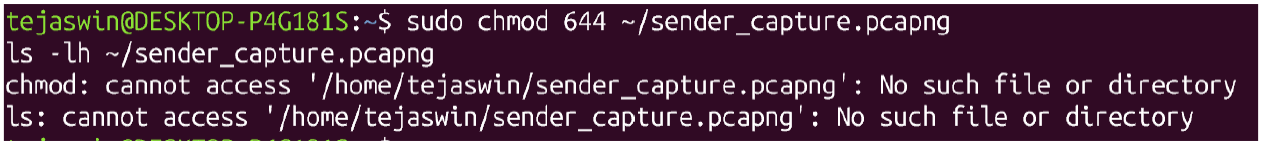



*Figure: chmod and ls errors*

## 6. Display filters: tcp, udp, icmp, tcp.len > 0

```
tshark -r ~/sender_capture.pcapng -Y "tcp" | head -n 15
tshark -r ~/sender_capture.pcapng -Y "udp" | head -n 15
tshark -r ~/sender_capture.pcapng -Y "icmp"
tshark -r ~/sender_capture.pcapng -Y "tcp.len > 0" | head -n 15
```

**Observation:** Observed in the filter output: TCP to port 443 with TLSv1.3 application data, UDP DNS queries, and ICMP echo requests and replies to 8.8.8.8.

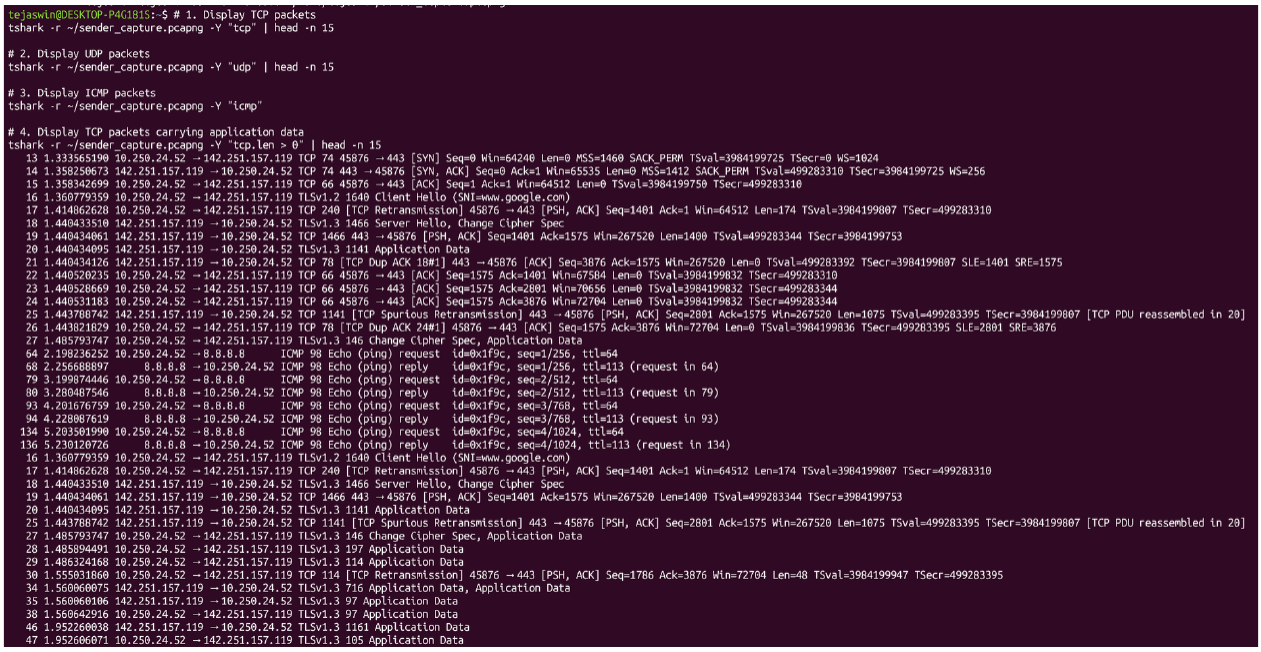



*Figure: Display filter output for tcp, udp, icmp and tcp.len > 0*

## 7. Header field extraction

```
tshark -r ~/sender_capture.pcapng -Y "tcp.len > 0" -c 1 -V | head -n 45
tshark -r ~/sender_capture.pcapng -Y "udp" -c 1 -T fields \
-e frame.len -e eth.src -e eth.dst -e ip.hdr_len -e ip.len -e udp.length
```

| Field | Value | Source |
|---|---|---|
| Ethernet II header | 14 bytes | Protocol standard (reference, not measured) |
| IPv4 header (no options) | 20 bytes | Protocol standard (reference, not measured) |
| Sample frame and UDP length values | Not legible in screenshot | Cut-off terminal output |

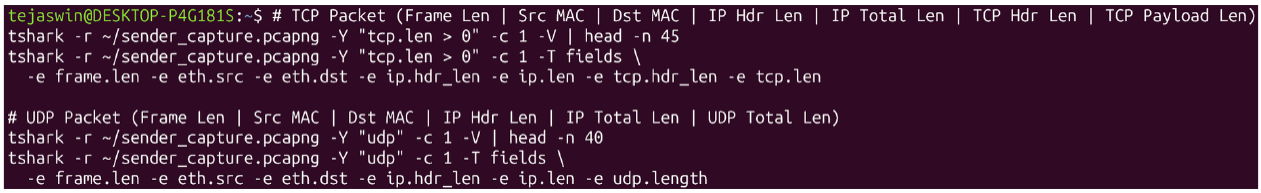



*Figure: Header field extraction commands*